VTSMOC: An Efficient Voronoi Tree Search Boosted Multiobjective Bayesian Optimization With Constraints for High-Dimensional Analog Circuit Synthesis

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***Abstract*—Optimizing multiple competitive black-box objectives with tight constraints poses a common challenge in analog circuit design. Multiobjective Bayesian optimization (MOBO) is a sample-efficient approach to identify the optimal trade-offs, namely, the Pareto front (PF). However, existing MOBO methods exhibit limitations in handling high-dimensional design space, large sample budgets, many objectives and tight constraints. This article introduces VTSMOC, a sample-efficient and computationally lightweight approach for addressing high-dimensional constrained multiobjective optimization problems. VTSMOC decomposes the design space into Voronoi cells, dynamically constructing a hierarchical Voronoi tree through clustering observations with dominance relationships. Promising leaf nodes in the Voronoi tree are pinpointed by traversing the tree with gradient bandit. The diversity of PF is ensured by parallel sampling within different promising cells, selected using a diffusive strategy. We also propose the expected PF improvement (EPFI) and probability of PF improvement (PPFI) acquisition functions to facilitate the PF efficiently along the radial direction of PF surface. Compared to state-of-the-art methods, VTSMOC achieves significant improvements in both sample and computational efficiency.**

***Index Terms*—Constrained multiobjective Bayesian optimization (MOBO), expected Pareto front improvement (EPFI), high-dimensional optimization, Voronoi tree search.**

Manuscript received 23 April 2024; revised 23 July 2024; accepted 2 September 2024. Date of publication 6 September 2024; date of current version 21 February 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2020YFA0711900 and Grant 2020YFA0711901; and in part by the National Natural Science Foundation of China (NSFC) Research Projects under Grant 62141407, Grant 62304052, Grant 62474050, Grant 62474051, and Grant 92373207. This article was recommended by Associate Editor W. Yu. *(Corresponding authors: Zhaori Bi; Xuan Zeng.)*

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Digital Object Identifier 10.1109/TCAD.2024.3455932

I. Introduction

THE DESIGN of analog circuits presents significant challenges due to its labor-intensive and knowledge-based nature, creating bottlenecks in the product development cycle for mixed-signal electronic systems. Developing efficient optimization methods for analog circuit synthesis is crucial for advancing high-performance analog and mixed-signal circuit design automation. The primary stages of automated analog circuit synthesis encompass topology selection, device sizing, and layout generation. In this article, we focus on the device sizing problem within predetermined circuit topologies.

The optimization approaches for analog circuit sizing predominantly comprise model-based and simulation-based methods. Model-based methods employ pre-established models to approximate the landscape of circuit performance and conduct optimization based on these models. Typically, circuit models take the form of analytical equations derived from human knowledge [1], [2] or regression models constructed by fitting simulated datasets [3], [4]. However, characterizing the heterogeneous circuit behavior is challenging when considering the parasitic effects. Simulation-based methods consider the entire circuit system as a black-box function output by circuit simulators, with the search procedure guided by global optimization algorithms. Commonly used optimization algorithms include multistart point (MSP)-driven local search methods [5] or evolutionary algorithms (EAs), such as particle swarm optimization (PSO) [6], differential evolution (DE) [7], etc. However, the derivative approximations or genetic operations of these methods necessitate numerous time-consuming circuit simulations, leading to a slow convergence rate.

Recently, Bayesian optimization (BO) has emerged as an effective approach for optimizing costly analog circuits [8], [9]. BO utilizes Gaussian process (GP) model to capture the circuit behavior and generates the next query candidate informed by acquisition functions (AFs), which makes BO a sample-efficient method for low-dimensional single objective optimization (SOO) problems with fewer than 20 design variables, whereas practical circuit optimization problems often involve dozens to hundreds of design variables,

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multiple competitive objectives and tight constraints. Hence, the significance of high-dimensional (HD) multiobjective BO (MOBO) is particularly pronounced. Developing efficient high-dimensional MOBO methods mainly grapples with three pivotal challenges.

1) *High-Dimensional Design Space:* As the design space expands geometrically with variable dimensionality, sparse observations in the HD space lead to large posterior uncertainty of the GP model. Consequently, the guidance provided by the AF in HD space deteriorates significantly, resulting in blind exploration of the entire space.
2) *Many Objectives and Constraints:* Most existing multiobjective AFs draw inspiration from hypervolume or entropy search (ES). The computational complexity of hypervolume escalates super-exponentially with the number of objectives, limiting its practicality for many-objective problems [10]. ES-based AFs lack explicit computational expressions, relying instead on Monte Carlo (MC) integration, which is computationally intractable and suffers from poor numerical precision. Furthermore, the MC integration-based AFs encounter challenges with tight constraints, as the feasible region is scarcely reached through MC sampling [11], [12].
3) *Large Sample Budgets:* The time bottleneck in MOBO arises from the training of GP models and the optimization of multiobjective AFs. The complexity of GP training scales cubically with the number of observations N. Commonly used AFs, such as MESMO [12] and EHVI [11], require time-consuming numerical approximation and are limited to small sample budgets (typically $N \leq 200$), which yields meagre success for HD heterogeneous problems.

High-dimensional Bayesian optimization (HDBO) strives to extend BO to HD design space. Core strategies encompass *dimension decoupling*, *subspace embedding*, and *region restriction*. *Dimension decoupling* methods aim to discern the additive structure of objective function with additive GP models and decouple the HD space into disjoint or overlapping low-dimensional subspaces [13], [14]. However, training a collection of GPs is computationally expensive and impractical for large sample budgets. *Subspace embedding* methods cast the HD problem into effective low-dimensional subspace through random embedding [15] or sensitive variable selection [16], [17]. Unfortunately, the presumption of intrinsic low dimensionality often does not hold for real-world circuit optimization problems. *Region restriction* is an efficacious approach that directly tackles the HD space. By conducting local BO within promising local domains, this approach significantly mitigates the blind exploration spanning the entire space and achieves higher-sample efficiency. Prominent works include the explicit rectangular local domains in TuRBO [18] and D3PBO [19], as well as the implicit irregular local domains in LA-MCTS [20] and VTS [21].

The geometric attributes of the target local domain directly influence the performance of *region restriction*-based HDBO methods. An ideal local domain boundary should be *irregular*, *convex*, *cheap-to-evaluate*, *HD-scalable*, and *fine-grained*. Irregular boundaries adapt better to heterogeneous function landscapes, while convex boundaries facilitate the utilization of advanced optimization algorithms to maximize AFs. HD-scalability is also crucial, as sparse observations in HD space easily lead to overfit boundaries, thereby misleading the search process, as exemplified by the SVM boundary in LA-MCTS [20]. Moreover, fine-grained local domains ensure that each observation contributes to updating the geometry of the local domain and guiding the search direction. Currently, VTS [21] is the sole method that provides fine-grained domain decomposition. The properties of local domain for region restriction methods are summarized in Table I. Notably, the aforementioned efforts primarily focus on SOO problems.

TABLE I
PROPERTIES OF LOCAL DOMAIN FOR REGION RESTRICTION-BASED HDBO METHODS

Attributes	TuRBO [18]	LA-MCTS [20]	D3PBO [19]	VTS [21]
Irregular	✗	✓	✗	✓
Convex	✓	✗	✓	✓
Cheap-to-evaluate	✓	✗	✓	✓
HD-scalable	✓	✗	✓	✓
Fine-grained	✗	✗	✗	✓

MOBO aims to propose multiobjective sampling criterion. Popular MOBO methods can be classified into three categories: 1) *hypervolume improvement* (HVI)-based; 2) *entropy search*-based; and 3) *decomposition*-based methods. *HVI*-based methods attempt to expand the hypervolume (HV) of the region dominated by the Pareto set (PS). Expected HVI (EHVI) [11] generalizes the expected improvement (EI) to multiobjective optimization (MOO) problems using HV indicator. *ES*-based methods maximize the mutual information gain about the PS or Pareto front (PF) [12]. However, the intractable entropy computation involves Monte Carlo approximations, making it time-consuming and inaccurate. *Decomposition*-based MOBO methods, such as parEGO [22], disassemble MOO problems into a collection of SOO problems using random augmented Chebyshev scalarizations and optimize each subproblem simultaneously through BO.

Only a few works consider both multiobjectives and HD design space concurrently. Notable among them are the multiobjective extensions of TuRBO, i.e., LoCoMOBO [23] and MORBO [24]. LoCoMOBO employs Thompson sampling and NSGAII [25] to derive the PF from a series of sample functions and generate the next query points maximizing the HVI. MORBO combines trust region with the EHVI [11] acquisition function, achieving commendable sample efficiency for HD multiobjective problems. However, the complexity associated with HVI renders these methods iteratively sluggish. Furthermore, the sample-based AFs encounter challenges with tight black-box constraints. Hence, prevalent methods for high-dimensional constrained multiobjective optimization (CMOO) problems still revolve around multiobjective evolutionary algorithms (MOEAs) like NSGAII [25] and NSGAIII [26], despite their low-sample efficiency in genetic operations.

To extend BO to high-dimensional CMOO problems, we propose VTSMOC, a sample-efficient and computationally lightweight MOBO for CMOO problems spanning dozens to hundreds of dimensions. VTSMOC addresses HD design space challenges by decomposing it into polyhedral Voronoi

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cells and recursively classifying cells into "good" and "bad" clusters, forming a hierarchical Voronoi tree. For a diverse PF, a batch of candidates is generated from different promising cells selected with a diffusive strategy in the objective space. To achieve high sample and computation efficiency, we propose expected PF improvement (EPFI) and probability of PF improvement (PPFI) local sampling criteria, with explicit formulas derived from the joint GP model, facilitating the PF surface along the radial direction. The contributions of this article are summarized as follow.

1) We propose a novel Voronoi tree structure for high-dimensional domain decomposition to efficiently manage the large sample budgets.
2) We redesign the path selection, branch expansion, and reward back-propagation operations for CMOO problems. By recursively clustering cells based on dominance relationships, we construct a Voronoi tree to limit the maximal number of observations in each leaf node. The promising regions are identified by traversing the Voronoi tree from root to a target leaf node via gradient bandit.
3) We propose a batch selection strategy to preserve the diversity of PF. We cluster Voronoi cells in the objective space and select the cell with maximal hypervolume contribution (HVC) from each cluster for further exploitation.
4) We present novel sample-efficient local sampling criteria, i.e., EPFI and PPFI, explicitly derived from joint GP model. The black-box constraints are addressed through the Probability of Feasibility (PoF).

We evaluate the performance of VTSMOC with both synthetic functions and practical analog circuits. The experimental results demonstrate that VTSMOC exhibits prominent sample-efficiency and reduced computation cost, along with diverse PF for high-dimensional CMOO problems. A Python implementation of VTSMOC can be accessed at https://github.com/adzhao072/VTSMOC.

II. Background

A. Problem Formulation

We formulate the analog circuit sizing problem as a constrained multiobjective optimization problem as follows:

$$\begin{aligned} &\min_{\boldsymbol{x}\in\mathcal{X}} \ \boldsymbol{f} = \left[f_1(\boldsymbol{x}), \ldots, f_M(\boldsymbol{x})\right]^T. \\ &\text{s.t.} \ \ g_i(\boldsymbol{x}) \le 0, \ \ i = 1, \ldots, n_g. \end{aligned} \tag{1}$$

Here, $\mathcal{X} = [0,1]^D$ represents the normalized design space. $\boldsymbol{x}$ denotes the design variable of the device sizes or input bias in D-dimensional space. $f_m(\boldsymbol{x})$ is the mth objective and $g_i(\boldsymbol{x})$ is the ith constraint, $(m \in \{1, \ldots, M\}, \ j \in \{1, \ldots, n_g\})$. $f_m(\boldsymbol{x})$ and $g_i(\boldsymbol{x})$ represent the circuit performance evaluated by circuit simulators. We assume that the objectives and constraints are all independent in this article. An observation refers to a sample $(\boldsymbol{x}, \boldsymbol{f}(\boldsymbol{x}), \boldsymbol{g}(\boldsymbol{x}))$ obtained from circuit simulations.

We first give the fundamental definitions for unconstrained MOO problems.

Definition 1 (Dominance Principle): For any $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{X}$, we say that $\boldsymbol{x}_1$ dominates $\boldsymbol{x}_2$, denoted as $\boldsymbol{x}_1 \prec \boldsymbol{x}_2$, if and only if for any $m \in \{1, \ldots, M\}$, $f_m(\boldsymbol{x}_1) \le f_m(\boldsymbol{x}_2)$ holds, and there exists $j \in \{1, \ldots, M\}$ such that $f_j(\boldsymbol{x}_1) < f_j(\boldsymbol{x}_2)$.

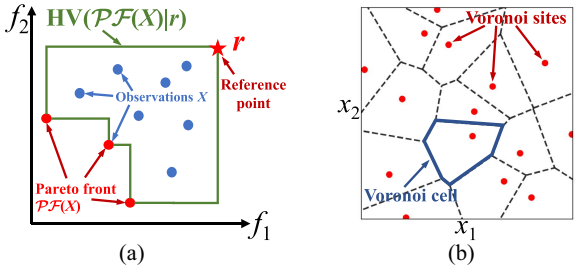


Fig. 1. Hypervolume (left) and Voronoi diagram (right). (a) Hypervolume. (b) Voronoi domain decomposition.

Definition 2 (Pareto Set and PF): Suppose $X \subset \mathcal{X}$ is an observed dataset. The Pareto set is defined as

$$\mathcal{PS}(X) = \left\{\boldsymbol{x}_i \in X | \nexists \boldsymbol{x}_j \in X, \text{s.t. } \boldsymbol{x}_j \prec \boldsymbol{x}_i\right\}. \tag{2}$$

The PF is defined as

$$\mathcal{PF}(X) = \{\boldsymbol{f}(\boldsymbol{x}) | \boldsymbol{x} \in \mathcal{PS}(X)\}. \tag{3}$$

For the constrained problems, we use the constraint violation (CV) to quantify the extent to which a solution $\boldsymbol{x}$ breaches the black-box constraints. The CV is defined as $\text{CV}(\boldsymbol{x}) = \sum_{i=1}^{n_g} \max(0, g_i(\boldsymbol{x}))$. Then we extend the previous concepts to CMOO problems.

Definition 3 (Constrained Dominance Principle): For any $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{X}$, we say that $\boldsymbol{x}_1$ constrainedly dominates $\boldsymbol{x}_2$, denoted as $\boldsymbol{x}_1 \prec_c \boldsymbol{x}_2$, under one of the following conditions:

$$\begin{cases} \boldsymbol{f}(\boldsymbol{x}_1) \prec \boldsymbol{f}(\boldsymbol{x}_2), \ \text{if CV}(\boldsymbol{x}_1) = 0 \text{ and CV}(\boldsymbol{x}_2) = 0 \\ \text{CV}(\boldsymbol{x}_1) = 0 \text{ and CV}(\boldsymbol{x}_2) \ne 0 \\ \text{CV}(\boldsymbol{x}_1) < \text{CV}(\boldsymbol{x}_2), \ \text{if CV}(\boldsymbol{x}_1) \ne 0 \text{ and CV}(\boldsymbol{x}_2) \ne 0. \end{cases} \tag{4}$$

Definition 4 (Constrained Pareto Set and PF): Given an observed dataset $X \subset \mathcal{X}$, the constrained Pareto set is defined as

$$\mathcal{PS}_c(X) = \left\{\boldsymbol{x}_i \in X | \text{CV}(\boldsymbol{x}_i) = 0 \text{ and } \nexists \boldsymbol{x}_j \in X, \text{s.t. } \boldsymbol{x}_j \prec_c \boldsymbol{x}_i\right\}. \tag{5}$$

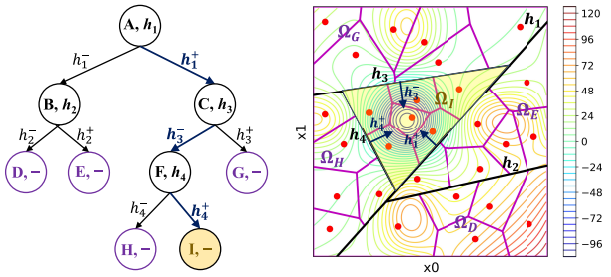
The constrained PF is defined as

$$\mathcal{PF}_c(X) = \{\boldsymbol{f}(\boldsymbol{x}) | \boldsymbol{x} \in \mathcal{PS}_c(X)\}. \tag{6}$$

The hypervolume (HV) indicator is a commonly used metric to evaluate the quality of a PF, which can assess both the convergence performance and diversity of the PF. Definition 5 precisely defines HV for multiobjective problems. Fig. 1(a) illustrates the HV indicator for the two-objective optimization problem min $[f_1, f_2]^T$. The red star represents the reference point $\boldsymbol{r}$. Notably, the computation complexity of HV grows exponentially with the number of objectives M. Specifically, HV computation using the widely used box decomposition method has a complexity of $\mathcal{O}(n^{\lfloor M/2 \rfloor + 1})$, where n is the number of nondominated solutions [27].

Definition 5 (Hypervolume Indicator): Given an observed dataset $X \subset \mathcal{X}$ and a reference point $\boldsymbol{r} \in \mathbb{R}^M$, the hypervolume of X is the M-dimensional Lebesgue measure of the region dominated by $\mathcal{PF}(X)$ (or $\mathcal{PF}_c(X)$) and bounded from above by $\boldsymbol{r}$.

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(a) (b)

Fig. 2. Depiction of the proposed Voronoi tree structure. The search space $\mathcal{X}$ is recursively partitioned by a sequence of hyperplanes $\mathcal{H} = \{h_1, h_2, h_3, h_4\}$, resulting in a BSP tree (left). Each leaf node represents a polytope-bounded Voronoi nest (right). The complete set of leaf nodes forms a *partition* of $\mathcal{X} \coloneqq \Omega_D \cup \Omega_E \cup \Omega_G \cup \Omega_H \cup \Omega_I$. The delineation of the boundary of a target leaf node Ω_I is determined by the blue path $A \to C \to F \to I$. (a) Voronoi tree and (b) corresponding design space partition.

B. Voronoi Domain Decomposition

Voronoi diagram is a popular method in computational geometry that decomposes the design space with convex polyhedrons. Consider N observations $X = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$ in $\mathcal{X}$, the Voronoi diagram meshes $\mathcal{X}$ into N Voronoi cells, which is defined in (7) [29]. The observation $\boldsymbol{x}_j$ is referred to as the "site" of Voronoi cell $\mathrm{Vor}(\boldsymbol{x}_j)$

$$\mathrm{Vor}(\boldsymbol{x}_j) = \left\{\boldsymbol{x} \in \mathcal{X} | \forall \boldsymbol{x}_i \in X, ||\boldsymbol{x} - \boldsymbol{x}_j|| \leq ||\boldsymbol{x} - \boldsymbol{x}_i||\right\} \tag{7}$$

where $||\cdot||$ represents the Euclidean distance. The Voronoi diagram $\mathrm{Vor}(X)$ establishes a complete partition of the design space $\mathcal{X}$ inspired by the observations X. Fig. 1(b) shows an illustration of the Voronoi domain decomposition.

III. Proposed Voronoi Tree Structure

Solving high-dimensional problems typically requires a large number of observations. The "divide and conquer" strategy efficiently handles large sample budgets by classifying observations and simultaneously decomposing the HD design space $\mathcal{X}$ into separate local domains. Confining BO within promising local domains can significantly improve the sample efficiency and reduce computational burden. In this section, we propose a novel Voronoi tree structure for high-dimensional domain decomposition, which is a variant of the binary space partition (BSP) tree tailored to the Voronoi diagram. Benefiting from the hierarchical Voronoi tree structure, we achieve efficient querying and assessment of numerous local domains.

The hierarchical domain decomposition method based on the Voronoi tree structure recursively partitions the design space $\mathcal{X}$ using a sequence of hyperplanes $\mathcal{H} = \{h_1, \ldots\}$. Fig. 2 provides an illustration of the Voronoi tree structure. Consider an observed dataset $X \subset \mathcal{X}$. The Voronoi tree $\mathcal{T}$ is specified by the hyperplanes $\mathcal{H}$ and observations X. Each node represents a convex polyhedral subdomain within the design space $\mathcal{X}$, with the root node A representing the entire space $\mathcal{X}$. Each nonleaf node corresponds to a partition hyperplane h_i with the explicit linear expression

$$h_i \coloneqq \left\{\boldsymbol{x} \in \mathcal{X} | \boldsymbol{w}_{h_i}^T \boldsymbol{x} + w_{0,h_i} = 0\right\}. \tag{8}$$

Its two extending edges correspond to the positive and negative half-spaces, h_i^+ and h_i^-, divided by h_i. Specifically, the positive half-space h_i^+ is defined as

$$h_i^+ \coloneqq \left\{\boldsymbol{x} \in \mathcal{X} | \boldsymbol{w}_{h_i}^T \boldsymbol{x} + w_{0,h_i} > 0\right\}. \tag{9}$$

And the negative half-space h_i^- is defined as

$$h_i^- \coloneqq \left\{\boldsymbol{x} \in \mathcal{X} | \boldsymbol{w}_{h_i}^T \boldsymbol{x} + w_{0,h_i} < 0\right\}. \tag{10}$$

The path from the root to a leaf node determines the boundary of the leaf node in the design space. For example, in Fig. 2, the blue path $A \xrightarrow{h_1^+} C \xrightarrow{h_3^-} F \xrightarrow{h_4^+} I$ in the Voronoi tree $\mathcal{T}$ identifies the boundary of $\Omega_I \coloneqq \mathcal{X} \cap h_1^+ \cap h_3^- \cap h_4^+$. Each leaf node v_{leaf} represents a *Voronoi nest*, a polytope-bounded Voronoi diagram induced by the observations located within $\Omega_{v_{\text{leaf}}}$, denoted as $\mathrm{Vor}(X \cap \Omega_{v_{\text{leaf}}})$. The entirety of leaf nodes constitutes a *partition* of the design space $\mathcal{X}$.

As the number of observations increases during the optimization process, the Voronoi tree $\mathcal{T}$ dynamically expands to accommodate the large sample budgets. This dynamic expansion limits the maximal number of observations within each leaf node, significantly reducing the computational complexity of local BO. Additionally, the size of promising Voronoi nest is gradually refined, leading to higher-sample efficiency in high-dimensional design space.

The partition hyperplanes $\mathcal{H}$ can be conveniently stored in the nonleaf nodes of the Voronoi tree $\mathcal{T}$. However, determining the explicit boundary of each Voronoi cell within the Voronoi nest is impractical in terms of both storage and computation, especially in HD design space. Therefore, we adopt an implicit Voronoi boundary determination method based on the property described in (7). Without loss of generality, we assume a selected promising Voronoi cell $\mathrm{Vor}(\boldsymbol{x}_*)$ within the target Voronoi nest $\mathrm{Vor}(X_I)$, where $X_I = X \cap \Omega_I$. For a newly candidate $\boldsymbol{x} \in \mathcal{X}$, we first utilize the explicit hyperplanes along the path from root to the target leaf node I to evaluate whether $\boldsymbol{x}$ belongs to Ω_I. If $\boldsymbol{x} \in \Omega_I$, we then perform nearest neighbor search (NNS) to determine whether $\boldsymbol{x}_*$ is the nearest neighbor of $\boldsymbol{x}$ among the observed subset X_I. If this condition holds true, then $\boldsymbol{x}$ is considered to be within the promising cell $\mathrm{Vor}(\boldsymbol{x}_*)$. Otherwise, $\boldsymbol{x}$ lies outside of the promising cell $\mathrm{Vor}(\boldsymbol{x}_*)$ and is rejected. In this article, the nearest neighbor search is implemented by *k-d* tree.

IV. Proposed VTSMOC Method

A. General Framework of VTSMOC

The general framework of VTSMOC is depicted in Fig. 3, consisting of four key phases.

1) Domain Decomposition via Dynamic Voronoi Tree Construction: We follow the hierarchical structure of VTS [21] and redesign the strategies for path selection, branch expansion, and reward back-propagation for CMOO problems. The Voronoi tree $\mathcal{T}$ is a binary space partition tree of the search space $\mathcal{X}$. If the number of observations in a leaf node reaches the maximal node size N_{leaf}, the tree continues to grow by segregating the observations into "good" and "bad" clusters to efficiently manage the growing number of observations.

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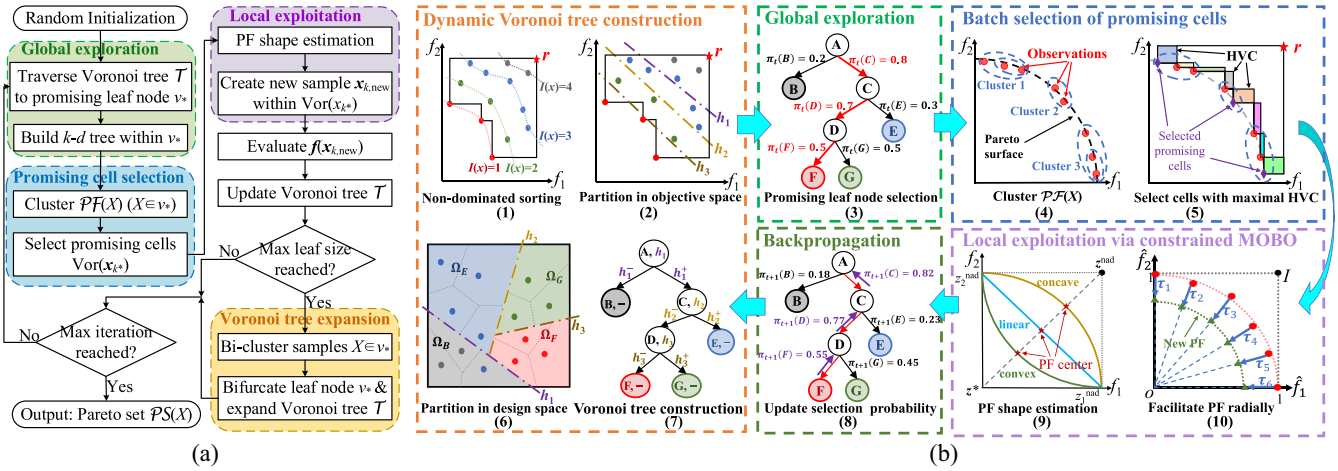


Fig. 3. Illustration of the search mechanism of VTSMOC. (a) Flowchart of the proposed VTSMOC algorithm. (b) Detailed illustration of VTSMOC. (b1)–(b2), (b6), and (b7) Dynamic Voronoi tree construction. The observations are clustered based on their nondominance rank to form polyhedral Voronoi nests with varying performance levels. The split operation corresponds to an expansion of the Voronoi tree $\mathcal{T}$, recursively partitioning the design space. (b3) Global exploration via Voronoi tree search. The target Voronoi nest is identified by traversing the Voronoi tree $\mathcal{T}$ from the root node to a target leaf node using gradient bandit. (b4) and (b5) Batch selection of promising Voronoi cells. Observations are clustered in the objective space through agglomerative hierarchical clustering. Promising Voronoi cells with maximal hypervolume improvement from each cluster are selected for further exploitation. (b8) Reward backpropagation. Voronoi tree $\mathcal{T}$ is updated with newly acquired observations. (b9) and (b10) Local exploitation via constrained MOBO. The PF is facilitated radially by maximizing the proposed EPFI or PPFI acquisition functions within the promising Voronoi cells. The tight constraints are tackled using the probability of feasibility.

2) Global Exploration via Voronoi Tree Search: The construction of Voronoi tree $\mathcal{T}$ converts the global exploration of the high-dimensional space into a tree search problem. In each iteration t, the promising region is identified by traversing from root to a promising leaf node v_*^t.

3) Batch Selection of Promising Cells: To achieve parallel sampling, we cluster the cells of the target node v_*^t in the objective space. From each cluster k, ($k \in \{1, \ldots, B\}$), we select the cell Vor($x_{k,*}$) with highest HVC for local exploitation.

4) Local Exploitation via Constrained MOBO: For local exploitation, we estimate the shape of PF surface, and employ local MOBO by maximizing the EPFI or probability of PPFI acquisition function within the selected cells Vor($x_{k,*}$). The black-box constraints are tackled by Probability of Feasibility (PoF). Finally, the objectives and constraints are evaluated through circuit simulations, and the Voronoi tree is updated through reward back-propagation. By iteratively performing these steps, VTSMOC attains high-sample efficiency and maintains low-computation cost for CMOO problems over high-dimensional spaces.

B. Dynamic Voronoi Tree Construction

The number of observations in each leaf node directly influences the computational complexity of training local GP models and maximizing the acquisition function. Therefore, if the number of observations in the target leaf node v_*^t reaches the maximum threshold N_{leaf}, we proceed to split the leaf node v_*^t by dynamic Voronoi tree construction.

Fig. 3(b1), (b2), (b6), and (b7) provides an illustration of the dynamic Voronoi tree construction for CMOO problems. To assess the quality of observations in multiobjective problems, we first rank the observations with fast nondominated sorting [25]. Observations with a lower-nondomination rank, denoted as $I(\boldsymbol{x})$, are considered more favorable than those with a higher $I(\boldsymbol{x})$. For observations in the Pareto set $\boldsymbol{x} \in \mathcal{PS}(X)$, we have $I(\boldsymbol{x}) = 1$. Then we cluster the observations, considering $[\boldsymbol{x}, I(\boldsymbol{x})]$, into "good" and "bad" clusters with K-means. The partition boundary h is determined by the perpendicular bisector of the cluster centroids. Each split operation corresponds to an expansion of the Voronoi tree $\mathcal{T}$, and the complete set of leaf nodes in Voronoi tree $\mathcal{T}$ represents a hierarchical partition of the design space $\mathcal{X}$. Through recursive subdivision of the leaf nodes, the high-performance region becomes progressively smaller, eventually converging to the true PF. For constrained problems, we apply the constrained dominance principle as defined in Definition 3 for nondominated sorting.

C. Global Exploration via Voronoi Tree Search

The PS of CMOO problems is an $M-1$ dimensional piecewise continuous manifold [30] in the design space. Unlike single objective optimization, where the global optimum exactly resides in a single promising leaf node v_*^t, the PS in CMOO problems may distribute across various leaf nodes within the Voronoi tree $\mathcal{T}$. The UCB-based branch selection criteria, as utilized in [21], tends to be a local method that repeatedly visits the same promising node during successive iterations. For an ill-conditioned scenario where $N \leq \exp(n_{\text{worse}}(\Delta Q/C_p)^2)$, the suboptimal node v_{worse} will never be visited during the total N observations. Here, ΔQ denotes the quality difference between the two children, and C_p is the exploration-exploitation balance. This potentially results in an anisotropic Pareto surface or overlooking portions of the PS that are distributed across other leaf nodes. Therefore, the UCB-based approach is less suitable for effectively discovering a diverse and isotropic PF [31].

In this work, we traverse the Voronoi tree $\mathcal{T}$ in a probabilistic manner. For each iteration t, a promising leaf node v_*^t is selected for exploitation by traversing from the root of $\mathcal{T}$. As illustrated in Fig. 3(b3), the red path $A \rightarrow C \rightarrow D \rightarrow F$

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Algorithm 1 Batch Selection of Promising Cells

Require: Observations $X_*^t \in v_*^t$, batch size B, reference point $\boldsymbol{r}$.

1: $\mathcal{C} = \{\{\boldsymbol{y}\},$ for $\boldsymbol{y} \in \mathcal{PF}(X_*^t)\}$, $N_{\text{cluster}} = |\mathcal{C}|$;
2: **while** $N_{\text{cluster}} > B$ **do** ▷ Hierarchical clustering
3: $d_{\min} = \infty$;
4: **for** $c_i, c_j \in \mathcal{C}$ and $c_i \neq c_j$ **do**
5: Compute the cluster distance $d(c_i, c_j)$ via Eq. (13);
6: **if** $d(c_i, c_j) < d_{\min}$ **then**
7: $P = (c_i, c_j)$, $d_{\min} = d(c_i, c_j)$;
8: **end if**
9: **end for**
10: Merge two clusters P, $N_{\text{cluster}} = N_{\text{cluster}} - 1$;
11: **end while**
12: **for** $k = 1$ to B **do** ▷ Promising cell selection
13: $\boldsymbol{x}_{k,*} = \arg\max_{\boldsymbol{x}_j \in c_k} \Delta_{\text{HV}}(\boldsymbol{x}_j, X_*^t | \boldsymbol{r})$;
14: **end for**
15: **return** Promising cells $\{\text{Vor}(\boldsymbol{x}_{k,*})\}$, $(k = 1, \ldots, B)$.

delineates the promising region Ω_F in the design space. The probability of selecting node v_i follows the Softmax function:

$$\pi_t(v_i) = \frac{e^{s_t(v_i)}}{\sum_{v_j \in \mathcal{S}(v_i)} e^{s_t(v_j)}} \tag{11}$$

where $\mathcal{S}(v_i)$ represents the set of child nodes of node v_i's parent. $s_t(v_i)$ denotes the preference of node v_i, which is updated through reward back-propagation [the purple path in Fig. 3(b8)]. The preference is dynamically adjusted using the gradient bandit algorithm [32] across iterations

$$s_{t+1}(v_i) = s_t(v_i) + \beta \cdot \left(R_t - \bar{R}_t\right) \cdot \left(\mathbb{1}_{\{v_i = v_*^t\}} - \pi_t(v_i)\right). \tag{12}$$

Here, the step size parameter $\beta > 0$ controls the learning rate. The reward R_t is set to 1 if the newly observed solution at iteration t belongs to the Pareto set. Otherwise, R_t is 0. $\bar{R}_t$ is the average reward over the previous t iterations. Initially, each child node is equally likely to be visited. Through the reinforcement learning procedure, the probability of visiting promising nodes containing Pareto solutions increases over time, while the probability of visiting suboptimal nodes gradually decreases.

D. Batch Selection of Promising Cells

We introduce a sample-efficient parallelism strategy in VTSMOC. To prevent intensive or repeated sampling within the same region, we generate a batch of B new observations from B different promising Voronoi cells. The batch selection approach for the target cells is detailed in Fig. 3(b4) and (b5) and Algorithm 1. Given the observations X_*^t within the promising node v_*^t, we first cluster $\mathcal{PF}(X_*^t)$ in the objective space into B clusters via agglomerative hierarchical clustering [33] (lines 2–11 in Algorithm 1). When merging clusters, our aim is to make the distribution of observations in each cluster more compact. Additionally, outliers that are far from other observations contribute more to the diversity of PF and should be optimized separately. Therefore, we compute the similarity of two clusters c_i and c_j $(i \neq j)$ using the maximal distance between observations

$$d(c_i, c_j) = \max_{\boldsymbol{y}_1 \in c_i, \boldsymbol{y}_2 \in c_j} ||\boldsymbol{y}_1 - \boldsymbol{y}_2||. \tag{13}$$

Subsequently, we select the cell $\text{Vor}(\boldsymbol{x}_{k,*})$ with maximal hypervolume contribution (HVC) from each cluster k for further local exploitation (lines 12–14 in Algorithm 1). The HVC of a sample $\boldsymbol{x}$ to a set of observations X is defined as [34]

$$\Delta_{\text{HV}}(\boldsymbol{x}, X | \boldsymbol{r}) = \text{HV}(\mathcal{PF}(X) | \boldsymbol{r}) - \text{HV}(\mathcal{PF}(X \setminus \boldsymbol{x}) | \boldsymbol{r}). \tag{14}$$

The parallel strategy in VTSMOC ensures the diversity and uniform coverage of the Pareto set in both design and objective space, which enables the algorithm to efficiently obtain a high-quality PF.

E. Local Exploitation via MOBO

Leveraging the proposed Voronoi domain decomposition mechanism and under the Lipchitz continuous assumption of the objective function, the objective function values within the promising cell $\text{Vor}(\boldsymbol{x}_{k,*})$ naturally tend to closely align with the site value $\boldsymbol{f}(\boldsymbol{x}_{k,*})$. To ensure the efficient advancement of the PF surface, we perform local MOBO along the radial direction of the PF surface. Additionally, we propose two computationally efficient multiobjective acquisition functions, i.e., EPFI and PPFI. The radial direction of the surface is determined by estimating the shape of the PF surface.

1) Pareto Surface Shape Estimation: For the CMOO problems with different metrics, the objective function is normalized by

$$\hat{f}_k = \frac{f_k - z_k^*}{z_k^{\text{nad}} - z_k^*}, k \in \{1, \ldots, M\} \tag{15}$$

where z_k^* is the infimum of f_k and z_k^{nad} is the supremum of f_k on the PF. We rely on the assumption that the normalized Pareto surface is related to a specific curve within the following family:

$$\sum_{k=1}^{M} \hat{f}_k^{\lambda} = 1, (\lambda > 0). \tag{16}$$

For $0 < \lambda < 1$, $\lambda = 1$, and $\lambda > 1$, the PF corresponds to a convex, linear, and concave shape, respectively, as illustrated in Fig. 4(a). $\boldsymbol{z}^*$ represents the *ideal point*, composed of the infimum on each objective. $\boldsymbol{z}^{\text{nad}}$ is the *nadir point*, composed of the supremum of each objective on the PF. The *PF center* is the intersection of the PF surface and the ideal-nadir line.

The shape of PF surface provides valuable guidance for the algorithm in determining the direction of PF advancement and improving sample efficiency. We use the Pareto set $\mathcal{PS}(X_*^t)$ to estimate the PF shape. The PF shape can be roughly estimated with the help of PF center C_{PF}. We define the ratio $\gamma = d_{OC_{\text{PF}}}/d_{\perp}$, where $d_{OC_{\text{PF}}}$ is the distance between origin O and PF center C_{PF}, and $d_{\perp} = \sqrt{M}/2$ is the distance from O to the hyper-plane $\sum_{k=1}^{M} \hat{f}_k = 1$. The shape of the PF surface can be determined based on the ratio γ [35]. For a linear PF, γ is approximately equal to 1. We use the existing Pareto solutions $\mathcal{PS}(X_*^t)$ to approximate the PF center C_{PF}. Consider a subset X_M selected from $\mathcal{PS}(X_*^t)$, which contains M Pareto solutions

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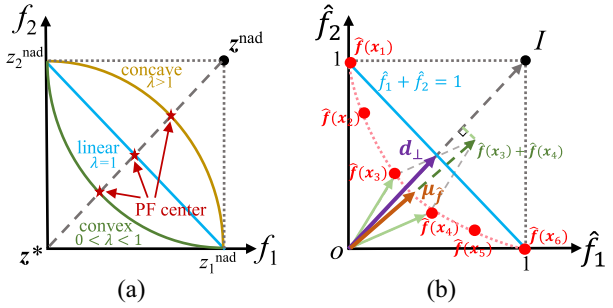


Fig. 4. Illustration of PF surface shape estimation. The PF is normalized by transferring the ideal point z^* to the origin O and the nadir point z^{nad} to I (left). The PF surface shape is identified by the coordinates of the approximated PF center (right). (a) Shape of PF surface. (b) Estimation of PF shape.

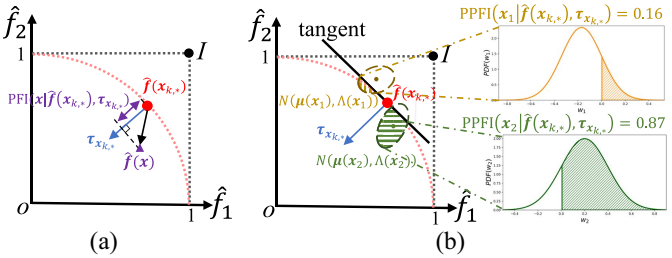


Fig. 5. Illustration of the PF improvement (left) and the proposed PPFI acquisition function (right). The GP predictive distribution for a newly candidate is represented by a joint Gaussian distribution, depicted as the ellipses in the right figure. The PPFI of a newly candidate is given by (22). (a) PF improvement. (b) PPFI.

forming the smallest angle with the M dimensional vector $I = [1, \ldots, 1]^T$. Then the PF center can be approximated by (17), as illustrated in Fig. 4(b)

$$\boldsymbol{\mu}_{\hat{f}} = \frac{1}{M} \sum_{\boldsymbol{x}_j \in X_M} \hat{\boldsymbol{f}}(\boldsymbol{x}_j). \tag{17}$$

The ratio γ is then approximated by

$$\gamma = \frac{d_{OC_{\text{PF}}}}{d_\perp} \approx \frac{||\boldsymbol{\mu}_{\hat{f}}||}{d_\perp} = \frac{2 \cdot ||\sum_{\boldsymbol{x}_j \in X_M} \hat{\boldsymbol{f}}(\boldsymbol{x}_j)||}{M\sqrt{M}}. \tag{18}$$

Thus, the estimation of Pareto surface shape is as follows: 1) if $||\gamma - 1|| \leq \epsilon$ or $|\mathcal{PS}(X_*^t)| \leq M$, PF is linear; 2) if $\gamma < 1 - \epsilon$, PF is convex; and 3) if $\gamma > 1 + \epsilon$, PF is concave. ϵ is the error tolerance and set to 0.1 in this article. In the early stages, when there are insufficient observations on the PF, i.e., $|\mathcal{PS}(X_*^t)| \leq M$, we do not have enough confidence to determine the PF shape. In such cases, we classify it as linear to maintain a consistent descent on different objectives. As the optimization progresses and more Pareto solutions are obtained, we can determine the PF shape accurately.

In this study, we focus on the elementary PF conforming to the curves family specified by (16). Our objective is to roughly determine the radial orientation of the PF surface and subsequently formulate multiobjective acquisition functions based on the PF surface's shape. For the intricate or complicated PFs, mature regression approaches can effectively capture their landscapes. Therefore, we refrain from further exploration into this aspect within the confines of this article.

2) Multiobjective Acquisition Function: We perform local MOBO concurrently in each promising cell $\text{Vor}(\boldsymbol{x}_{k,*})$ selected in Section IV-D. To enhance sample efficiency, we optimize the acquisition function along the radial direction $\boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}$ of the PF surface. The radial direction at a Pareto optimal solution $\boldsymbol{x}_{k,*}$ is given by

$$\tau(\boldsymbol{x}_{k,*}) = \begin{cases} \hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*}) - I, & \text{if PF is convex} \\ -I, & \text{if PF is linear} \\ -\hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*}), & \text{if PF is concave.} \end{cases} \tag{19}$$

By normalizing $\boldsymbol{x}_{k,*}$ into a unit vector $\boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}$, we can give the definition of PF improvement (PFI). For a newly observation $\boldsymbol{x}$, the PFI concerning a current Pareto optimal solution $\boldsymbol{x}_{k,*}$ is defined as

$$\text{PFI}\left(\boldsymbol{x}|\hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*}), \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}\right) = \max\left(\left(\hat{\boldsymbol{f}}(\boldsymbol{x}) - \hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*})\right)^T \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}, 0\right). \tag{20}$$

Please refer to Fig. 5(a) for details. Let $w = (\hat{\boldsymbol{f}}(\boldsymbol{x}) - \hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*}))^T \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}$. We assume that the normalized observation $\hat{\boldsymbol{f}}$ obeys a joint Gaussian distribution $N(\boldsymbol{\mu}, \Lambda)$, where $\boldsymbol{\mu}$ and Λ are, respectively, the mean vector and covariance matrix provided by the GP models. If all the objectives are independent, Λ is a diagonal matrix. Then the projection w also follows a Gaussian distribution $N(\mu_w, \sigma_w^2)$, where

$$\begin{cases} \mu_w = \left(\boldsymbol{\mu} - \hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*})\right)^T \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}} \\ \sigma_w^2 = \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}^T \Lambda \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}. \end{cases} \tag{21}$$

Following the concept of probability of improvement (PI) and expected improvement (EI), we introduce the definition of PPFI and EPFI acquisition function for multiobjective problems as follows:

$$\begin{aligned} \text{PPFI}\left(\boldsymbol{x}|\hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*}), \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}\right) &= \mathbb{P}\left(\text{PFI}\left(\boldsymbol{x}|\hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*}), \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}\right) > 0\right) \\ &= \int_0^{+\infty} p(w)\ \mathrm{d}w = \Phi\left(\frac{\mu_w}{\sigma_w}\right) \end{aligned} \tag{22}$$

$$\begin{aligned} \text{EPFI}\left(\boldsymbol{x}|\hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*}), \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}\right) &= \mathbb{E}\left[\text{PFI}\left(\boldsymbol{x}|\hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*}), \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}\right)\right] \\ &= \int_{\mathbb{R}^M} \text{PFI}\left(\boldsymbol{x}|\hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*}), \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}\right) \cdot p\left(\hat{\boldsymbol{f}}\right)\ \mathrm{d}\hat{\boldsymbol{f}} \\ &= \int_0^{+\infty} w \cdot p(w)\ \mathrm{d}w = \mu_w \Phi\left(\frac{\mu_w}{\sigma_w}\right) \\ &\quad + \sigma_w \phi\left(\frac{\mu_w}{\sigma_w}\right). \end{aligned} \tag{23}$$

$\phi(\cdot)$ and $\Phi(\cdot)$ respectively denote the probability density function and cumulative distribution function of a standard Gaussian distribution. Fig. 5(b) presents an illustration of the PPFI acquisition function, where the ellipse depicts the 3-σ range of the GP predictive distribution. Leveraging the Lipschitz continuity of the objective function $\hat{\boldsymbol{f}}$, the objective function value within the target cell $\text{Vor}(\boldsymbol{x}_{k,*})$ closely aligns with the site observation $\hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*})$. Candidates with higher-PPFI values, such as $\boldsymbol{x}_2$ in Fig. 5(b), have a higher probability of advancing the PF.

For the CMOO problem with black-box constraints, we utilize the Probability of Feasibility (PoF) to penalize the infeasible regions where the constraints are not satisfied. The PoF for a candidate $\boldsymbol{x}$ is defined as

$$\text{PoF}(\boldsymbol{x}) = \mathbb{P}(\boldsymbol{g}(\boldsymbol{x}) \leq 0) = \prod_{i=1}^{n_g} \Phi\left(-\frac{\mu_{g_i}(\boldsymbol{x})}{\sigma_{g_i}(\boldsymbol{x})}\right) \tag{24}$$

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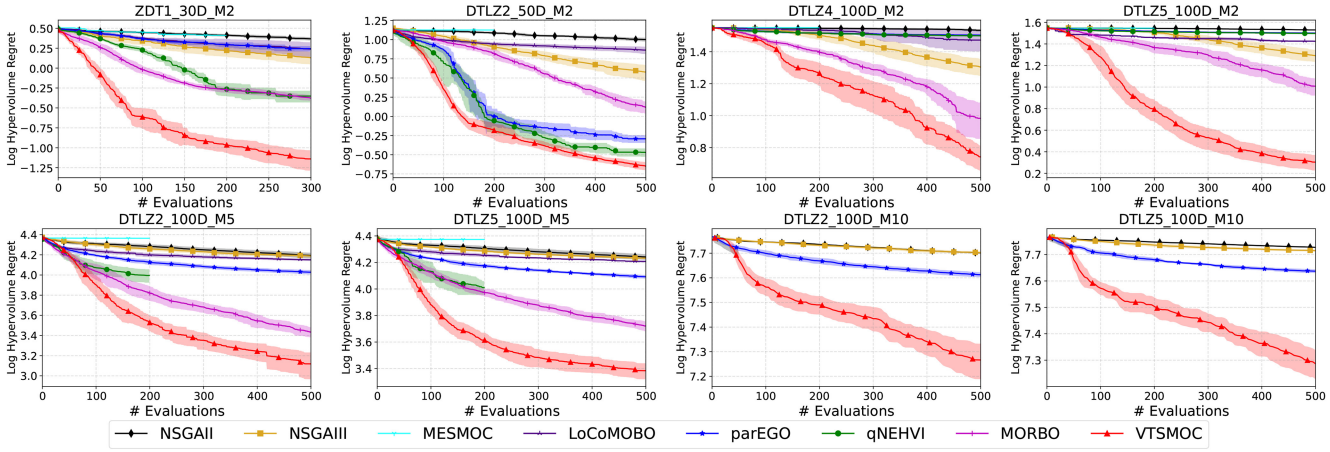


Fig. 6. Optimization results for high-dimensional multiobjective synthetic benchmarks. The logarithmic hypervolume regret is defined as $\log R(\mathcal{PF}(X)) = \log(\mathrm{HV}(\mathcal{PF}^*) - \mathrm{HV}(\mathcal{PF}(X)))$, where $\mathcal{PF}^*$ is the true PF.

where $\mu_{g_i}(\boldsymbol{x})$ and $\sigma_{g_i}(\boldsymbol{x})$ are, respectively, the predictive mean and standard deviation for the ith constraint of the local GP model. In the initial iterations, if the site value $\boldsymbol{g}(\boldsymbol{x}_{k,*})$ of the selected cell does not satisfy the constraints, we first optimize the PoF function to search for feasible regions. For the feasible region where $\boldsymbol{g}(\boldsymbol{x}_{k,*}) \leq 0$, we optimize the constrained multiobjective acquisition function

$$\alpha_c\left(\boldsymbol{x}|\hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*}), \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}\right) = \alpha\left(\boldsymbol{x}|\hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*}), \boldsymbol{\tau}_{\boldsymbol{x}_{k,*}}\right) \cdot \mathrm{PoF}(\boldsymbol{x}) \quad (25)$$

where $\alpha(\cdot)$ is the PPFI or EPFI acquisition function.

The proposed PPFI and EPFI acquisition functions are both in closed form of the predictive mean and variance of GPs, which can be calculated in $\mathcal{O}(M + n_g)$ time as shown in (21)–(25) with the given independent GP posteriors. These cheap-to-evaluate acquisition functions can be optimized efficiently by global optimization methods, such as CMA-ES [36] and MSP-BFGS [37]. VTSMOC is a variant of decomposition-based MOBO methods, providing implicit local adaptive decomposition. For each selected cell $\mathrm{Vor}(\boldsymbol{x}_{k,*})$, the PFI-based acquisition function decomposes the multiobjective function $\hat{\boldsymbol{f}}$ around $\hat{\boldsymbol{f}}(\boldsymbol{x}_{k,*})$ with $\boldsymbol{\tau}_{k,*}$ as the scalarization vector. Compared to existing decomposition-based MOBO methods, such as parEGO [22], VTSMOC introduces three significant improvements:

1) Oriented PF Advancement: VTSMOC strategically moves along the radial direction of the PF surface, which achieves higher-sample efficiency than parEGO's random scalarization vectors.

2) Lower-Computational Complexity: The PPFI and EPFI acquisition functions in VTSMOC are derived from joint GP models, contrasting with the resource-intensive GP models utilized in parEGO for each SOO subproblem. Furthermore, VTSMOC reduces computational complexity stemming from GPs through dynamic Voronoi tree construction, enhancing scalability for high-dimensional scenarios with large sample budgets.

3) Handling Black-Box Constraints: VTSMOC introduces innovative features to effectively manage the black-box constraints.

F. Complexity Analysis

VTSMOC addresses the high-computational complexity of MOBO from two aspects: 1) we reduce the computational cost of GPs by recursively partitioning the space using Voronoi tree and training local GPs within the promising leaf node and 2) we present novel acquisition functions, PPFI and EPFI, which are in the closed form of GP posteriors and efficient to evaluate. The computational complexity of VTSMOC mainly comes from three components: 1) batch selection of promising cells; 2) training local GPs; and 3) optimization of acquisition functions. The batch selection of promising cells involves a $\mathcal{O}(MN_{\mathrm{leaf}}^2)$ complexity for fast nondominated sorting, and a $\mathcal{O}(N_{\mathrm{leaf}}^3)$ complexity for hierarchical clustering. VTSMOC maintains $M+n_g$ independent local GPs for each objective and constraint, resulting in a $\mathcal{O}((M+n_g)N_{\mathrm{leaf}}^3)$ GP training complexity. The optimization of acquisition function involves a $\mathcal{O}((M+n_g)N_{\mathrm{leaf}}^2)$ GP inference complexity and the complexity of NNS for implicit Voronoi cell boundary determination. The NNS implemented by k-d tree has a construction complexity of $\mathcal{O}(N_{\mathrm{leaf}}\log(N_{\mathrm{leaf}}))$ and a query complexity of $\mathcal{O}(\log(N_{\mathrm{leaf}}))$. Thus, the total complexity is

$$\mathcal{O}\Big((M+n_g)N_{\mathrm{leaf}}^3 + (M+n_g)N_{\mathrm{leaf}}^2 + N_{\mathrm{leaf}}\log(N_{\mathrm{leaf}}) + \log(N_{\mathrm{leaf}})\Big). \quad (26)$$

Since the maximal leaf size N_{leaf} is a constant parameter, VTSMOC maintains a consistent computational complexity regardless of the total number of observations N.

V. Experimental Results

We evaluate the performance of VTSMOC across a comprehensive benchmark suite, comprising ZDT, DTLZ and MW test problems. These benchmarks feature variations in the number of design variables (D), objectives (M), and different PF shapes. Additionally, we apply VTSMOC to optimize three practical analog circuits: 1) a charge pump; 2) an AC coupled instrumentation amplifier (ACCIA); and 3) an operational transconductance amplifier (OTA). All benchmarks are optimized in the high-dimensional space with $D \geq 20$ design variables.

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TABLE II
EXPERIMENTAL SETTINGS ON SYNTHETIC BENCHMARKS

Synthetic benchmarks	Dimensions D	Objectives M	Reference point $\boldsymbol{r}$
ZDT1_30D_M2	30	2	[1,4]
DTLZ2_50D_M2	50	2	[4,4]
DTLZ4_100D_M2	100	2	[6,6]
DTLZ5_100D_M2	100	2	[6,6]
DTLZ2_100D_M5	100	5	[8,8,8,8,8]
DTLZ5_100D_M5	100	5	[8,8,8,8,8]
DTLZ2_100D_M10	100	10	$[6]^{10}$
DTLZ5_100D_M10	100	10	$[6]^{10}$

We conduct a comparative analysis of VTSMOC against state-of-the-art MOBO methods, including hypervolume improvement-based approaches, such as MORBO [24], qNEHVI [11], and LoCoMOBO [23]; entropy search-based method MESMOC [12]; and decomposition-based method parEGO [22]. Additionally, we include commonly used MOEAs, such as dominance relationship-based NSGAII [25] and decomposition-based NSGAIII [26]. The experiments consist of 10 replications with 50 random initial points for each run. All experiments are executed on a Linux workstation with Intel Gold 6230 CPUs and 128-GB memory. For the parallel MOBO methods, including VTSMOC (this work), MORBO [24], qNEHVI [11], parEGO [22], and LoCoMOBO [23], we set the batch size $B = 5$.

A. Synthetic Benchmarks

The detailed experimental setups for the synthetic benchmarks are outlined in Table II. Recognizing the large computational cost of MESMOC associated with MC integration, we limit the function evaluations to $N = 200$ for MESMOC. In cases involving many objectives with $M = 5$, qNEHVI is also conducted with $N = 200$ evaluations, considering its hyper-exponential time complexity. For all other approaches, we assign an additional $N = 300$ function evaluations for ZDT1_30D_M2 and $N = 500$ function evaluations for the remaining benchmarks. We utilize the hypervolume indicator to evaluate the convergence and diversity of the obtained PFs.

The experimental results from 10 repeated runs are depicted in Fig. 6 and summarized in Table III. VTSMOC consistently outperforms alternative MOBO baselines, showcasing superior sample efficiency and markedly lower-computational costs. The entropy search-based MOBO method MESMOC struggles to handle the high-dimensional input space, exhibiting notably high-computational complexity and even poorer sample efficiency compared to NSGAII. For medium-sized problems with $D \leq 50$, qNEHVI also demonstrates notable improvement in hypervolume. As the design space expands to 100D, VTSMOC outperforms other baselines by a substantial margin, with MORBO trailing behind. In scenarios with many objectives, specifically $M \in \{5, 10\}$, VTSMOC continues to exhibit superior sample and computational efficiency. The decomposition-based MOBO method parEGO, lacking hypervolume computation and nondominated sorting operations, demonstrates modest computational costs for small sample budgets. qNEHVI also reports a concise runtime for the DTLZ4_100D_M2 and DTLZ5_100D_M2 benchmarks due to the scarcity of observations on the PF, which significantly reduces the computation cost for hypervolume. For the many-objective benchmarks with $M = 10$, the HVI-based methods MORBO, qNEHVI, and LoCoMOBO, as well as the entropy search method MESMOC, all fail to execute due to the extremely high-computational costs.

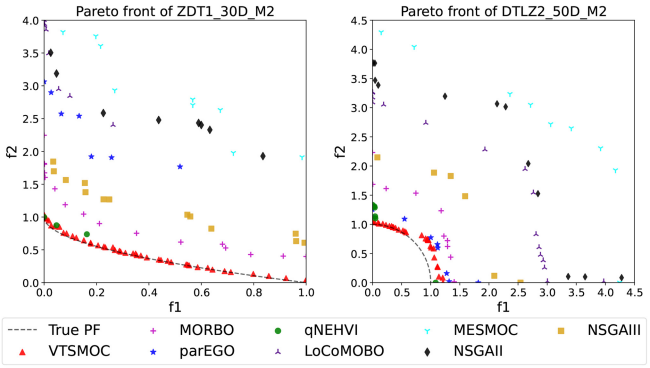


Fig. 7. PF of synthetic benchmarks. The PF corresponds to the median hypervolume across multiple runs.

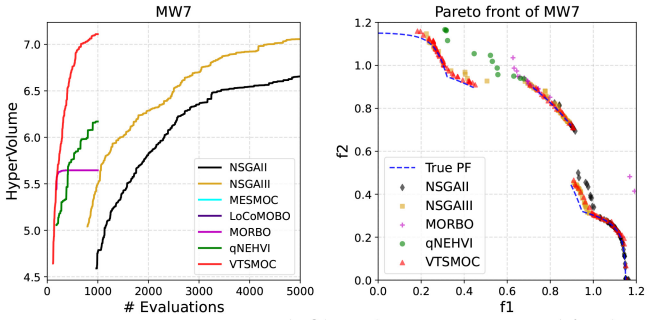


Fig. 8. Optimization results (left) and PF comparison (right) for MW7. The HV curves are plotted once all constraints are satisfied. The reference point for HV computation is $\boldsymbol{r} = [3, 3]$.

Fig. 7 presents a PF comparison between VTSMOC and other baselines for the ZDT1_30D_M2 and DTLZ2_50D_M2 benchmark. The ZDT1_30D_M2 benchmark exhibits a convex PF, whereas DTLZ2_50D_M2 corresponds to a concave PF. Remarkably, the PF obtained by VTSMOC not only encompasses a larger quantity of Pareto-optimal solutions but also demonstrates superior convergence and diversity. The comprehensive coverage of the PF by VTSMOC on the true PF verifies the effectiveness of the proposed tree traversal and batch selection strategy in discovering a diverse PF. Although qNEHVI shows a remarkable reduction in hypervolume regret, its PF is notably sparse and lacks diversity.

We also use the MW7 benchmark, which has 2 objectives and 2 constraints, to evaluate the performance of VTSMOC in handling complex and discontinuous PF. The constraints are difficult to satisfy, and the heterogeneous PF is challenging to approximate. We assign $N = 1000$ function evaluations to MOBO methods and $N = 5000$ function evaluations for the MOEAs. The experimental results are presented in Fig. 8. VTSMOC consistently outperforms other baselines with superior sample efficiency. Additionally, the PF of VTSMOC demonstrates better convergence and coverage in approximating the true PF compared to MOEAs, even with significantly fewer observations.

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TABLE III
EXPERIMENTAL RESULTS OF SYNTHETIC BENCHMARKS

Algorithms		VTSMOC	MORBO [24]	qNEHVI [11]	LoCoMOBO [23]	MESMO [12]	parEGO [22]	NSGAII [25]	NSGAIII [26]
ZDT1_30D_M2	HV	**3.59 ± 0.02**	3.24 ± 0.03	3.22 ± 0.07	1.93 ± 0.50	1.11 ± 0.04	1.91 ± 0.14	1.33 ± 0.11	2.32 ± 0.24
	Run time	**3.6min**	1.43h	31.7min	51.2min	13.55h	17.6min	< 1s	< 1s
DTLZ2_50D_M2	HV	**14.99 ± 0.03**	13.89 ± 0.25	14.87 ± 0.04	7.91 ± 1.06	1.87 ± 0.02	14.70 ± 0.06	5.18 ± 0.56	11.43 ± 0.90
	Run time	**7.9min**	2.21h	2.43h	1.61h	18.34h	10.8min	< 1s	< 1s
DTLZ4_100D_M2	HV	**29.96 ± 0.99**	25.85 ± 2.46	3.85 ± 0.33	5.77 ± 5.95	0.00 ± 0.00	3.57 ± 0.45	1.29 ± 0.61	15.31 ± 2.44
	Run time	**10.1min**	3.28h	15.7min	1.85h	16.73h	12.2min	< 1s	< 1s
DTLZ5_100D_M2	HV	**33.46 ± 0.33**	25.28 ± 1.85	4.02 ± 0.53	8.86 ± 0.49	0.00 ± 0.00	3.80 ± 0.48	1.60 ± 0.23	15.92 ± 2.61
	Run time	10.7min	3.49h	8.4min	1.83h	16.67h	**7.7min**	< 1s	< 1s
DTLZ2_100D_M5	HV($\times 10^4$)	**3.15 ± 0.05**	3.00 ± 0.03	2.28 ± 0.15	1.86 ± 0.05	0.96 ± 0.00	2.21 ± 0.06	1.71 ± 0.10	1.73 ± 0.06
	Run time	**16.3min**	11.49h	15.78h	9.93h	51.35h	17.1min	< 1s	< 1s
DTLZ5_100D_M5	HV($\times 10^4$)	**3.04 ± 0.04**	2.75 ± 0.05	2.25 ± 0.12	1.66 ± 0.03	0.91 ± 0.00	2.04 ± 0.04	1.53 ± 0.08	1.57 ± 0.03
	Run time	**28.6min**	12.32h	14.70h	9.94h	48.98h	31.4min	< 1s	< 1s
DTLZ2_100D_M10	HV($\times 10^7$)	**4.19 ± 0.49**	Fail	Fail	Fail	Fail	1.95 ± 0.94	1.01 ± 0.15	1.00 ± 0.06
	Run time	**32.0min**	> 50h	> 50h	> 50h	> 50h	1.45h	< 1s	< 1s
DTLZ5_100D_M10	HV($\times 10^7$)	**4.11 ± 0.37**	Fail	Fail	Fail	Fail	1.71 ± 0.07	0.71 ± 0.12	0.86 ± 0.05
	Run time	**27.9min**	> 50h	> 50h	> 50h	> 50h	1.35h	< 1s	< 1s

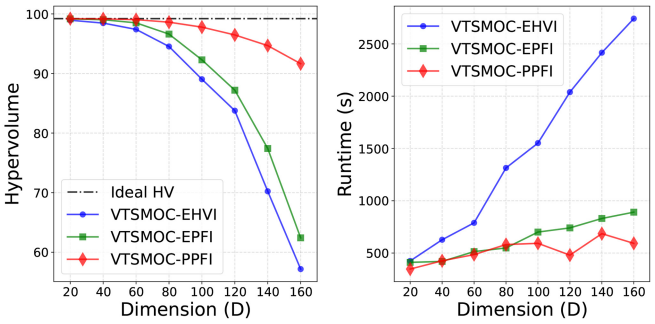


Fig. 9. Sensitivity analysis on acquisition functions and problem dimensions. The proposed PPFI acquisition function demonstrates better adaptation to high-dimensional spaces (left). The computational cost of proposed EPFI/PPFI acquisition function is significantly lower than EHVI (right).

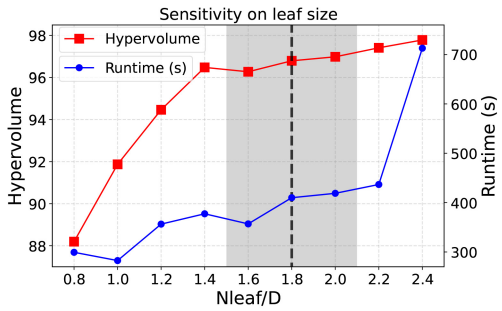


Fig. 10. Sensitivity analysis on leaf size N_{leaf}. A larger N_{leaf} leads to better hypervolume but incurs higher-time consumption.

B. Ablation Studies

VTSMOC provides superior sample and computation efficiency for high-dimensional MOBO. We conduct ablation studies to assess the algorithm's sensitivity to hyperparameters from three aspects: 1) the choice of acquisition functions; 2) various problem dimensions D; and 3) different leaf sizes N_{leaf}. The evaluation considers both convergence performance and runtime. We utilize the popular DTLZ2 benchmark with $M = 2$ objectives for our experiments. Each experiment is initiated with 50 random initial points, followed by a total of 500 additional function evaluations. The average results from 10 replicate runs are recorded. The reference point is defined as $\boldsymbol{r} = [10, 10]$ for hypervolume computation.

We evaluate the performance of VTSMOC incorporating various acquisition functions, including the proposed EPFI and PPFI, alongside the popular EHVI. The impact of problem dimensions D on the performance of different acquisition functions is depicted in Fig. 9. Notably, the performance of MOBO demonstrates exponential deterioration with the increase in the dimensionality of the design space. Both the proposed EPFI and PPFI acquisition functions exhibit higher-sample efficiency and significantly lower-computational costs compared to the popular EHVI method. Particularly for high-dimensional problems with $D \geq 100$, PPFI outperforms both EPFI and EHVI by a considerable margin. The runtime in the results increases linearly with the problem dimension D, primarily due to the computational complexity of the GP kernel, which typically scales linearly with the dimension.

Fig. 10 illustrates the impact of the maximal leaf size N_{leaf} on convergence and runtime of VTSMOC. A larger N_{leaf} provides a more accurate local GP model, leading to better-convergence results. However, beyond a certain range of N_{leaf}, the improvement in hypervolume by further increasing N_{leaf} becomes less evident. However, the computational complexity grows cubically with N_{leaf}, as indicated by (26). Therefore, setting a reasonable leaf size N_{leaf} within the range of $N_{\text{leaf}} \in [1.6D, 2.0D]$ typically achieves a good balance between convergence performance and computational cost.

C. Circuit Applications

We evaluate the performance of VTSMOC using three real-world analog circuits: 1) a charge pump circuit; 2) an OTA; and 3) an ACCIA. Each circuit comprises more than 30 design variables and presents significant challenges for multiobjective optimization algorithms due to stringent design constraints. Each method is executed 10 times with 50 random initial points. We assign $N = 1000$ circuit simulations for MOBO approaches and $N = 5000$ for MOEAs.

1) Charge Pump: The first benchmark circuit is a charge pump designed in the SMIC 40-nm process [5]. The circuit schematic and design metrics are illustrated in Fig. 11(a). This circuit consists of a total of 41 transistors, with 42 free design variables corresponding to the lengths and widths of the transistors, considering topological symmetry and current mirrors. For the optimization metrics, we consider both a two-objective scenario and a five-objective scenario.

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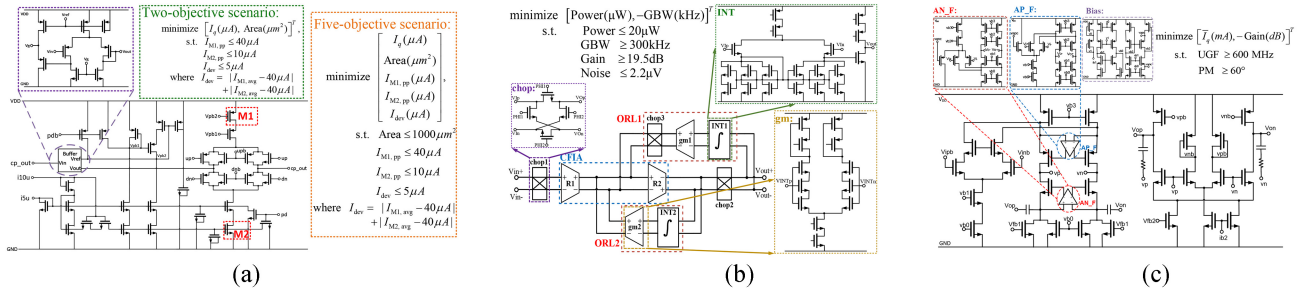


Fig. 11. Schematic and design specifications of benchmark circuits: (a) charge pump, (b) ACCIA, and (c) OTA.

TABLE IV
OPTIMIZATION RESULTS OF BENCHMARK CIRCUITS

Algorithms	#Simu	Charge pump ($M = 2$)			Charge pump ($M = 5$)			ACCIA			OTA		
		HV($\times 10^5$)	Runtime	#Succ	HV($\times 10^7$)	Runtime	#Succ	HV($\times 10^4$)	Runtime	#Succ	HV($\times 10^3$)	Runtime	#Succ
VTSMOC	1000	$\mathbf{1.81 \pm 0.06}$	**0.46h**	**10/10**	$\mathbf{4.90 \pm 0.63}$	**0.97h**	**10/10**	$\mathbf{1.72 \pm 0.08}$	**1.2h**	**10/10**	$\mathbf{1.05 \pm 0.04}$	**0.70h**	**10/10**
MORBO [24]	1000	Fail	8.9h	0/10	Fail	22.8h	0/10	0.15 ± 0.14	17.5h	9/10	0.41 ± 0.16	20.6h	4/10
qNEHVI [11]	1000	Fail	10.8h	0/10	Fail	3.2h	0/10	0.41 ± 0.08	20.6h	10/10	0.27 ± 0.23	14.3h	4/10
LoCoMOBO [23]	1000	Fail	7.0h	0/10	Fail	17.7h	0/10	0.15 ± 0.07	12.0h	9/10	0.42 ± 0.17	16.3h	4/10
MESMOC [12]	1000	0.28 ± 0.15	>50h	5/10	Fail	14.0h	0/10	0.43 ± 0.04	>50h	10/10	0.42 ± 0.15	>50h	9/10
NSGAII [25]	5000	1.62 ± 0.16	2.5h	10/10	4.25 ± 0.50	2.5h	10/10	1.61 ± 0.18	23.7h	10/10	1.04 ± 0.05	6.6h	10/10
NSGAIII [26]	5000	1.36 ± 0.30	2.6h	10/10	Fail	2.7h	0/10	1.42 ± 0.19	24.8h	10/10	0.90 ± 0.08	6.7h	10/10

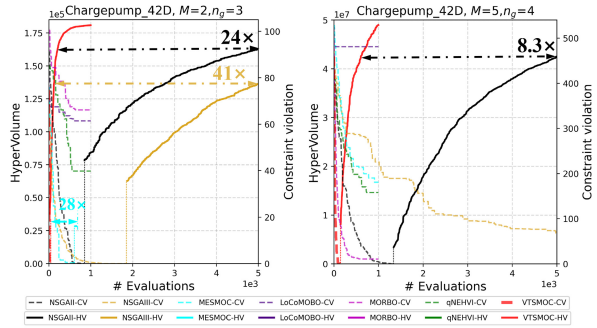


Fig. 12. Optimization results of the charge pump circuit, including two-objective scenario (left) and five-objective scenario (right). Tracked metrics include CV and HV. The HV (solid lines) are plotted after all the constraints are satisfied (CV = 0). For methods that fail to find the feasible region, the HV lines are absent.

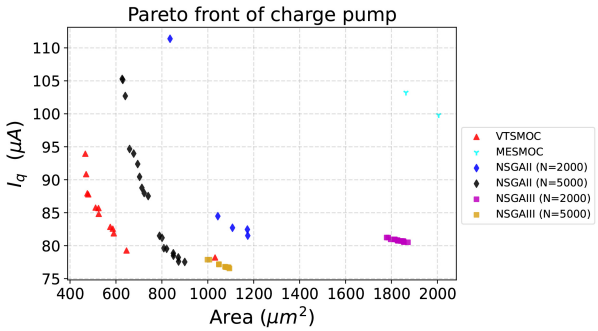


Fig. 13. PF of charge pump.

Two-Objective Scenario: The two-objective formulation includes 2 objectives and 3 constraints. The experimental results are presented in Fig. 12 and Table IV. The reference point selected for HV computation is $\boldsymbol{r} = [200(\mu A), 2000(\mu \text{m}^2)]$. As illustrated in Fig. 12, the reduction of constraint violation is depicted by the dashed curves. Once all the constraints are satisfied (CV = 0), the improvement of HV is traced with the solid curves. The absence of the HV curves for the HVI-based MOBO baselines, namely, qNEHVI, MORBO, and LoCoMOBO, is attributed to their inability to discover feasible solutions within the allocated 1000 circuit simulations. Remarkably, VTSMOC satisfies tight constraints with only a few dozen observations. Achieving equivalent HV, VTSMOC demonstrates a notable 24× and 41× speedup compared to NSGAII and NSGAIII, respectively, in terms of the number of circuit simulations. The HV curve of MESMOC is incomplete due to its iteration time exceeding 50 h. Similarly, VTSMOC is 28× faster than MESMOC for achieving comparable results in terms of circuit simulations. When considering runtime, VTSMOC achieves a 5.4× speedup compared to NSGAII while producing superior solutions.

Fig. 13 depicts a comparison of the obtained PFs, which correspond to the median HV. The charge pump circuit has a convex PF. The convergence of PF obtained by VTSMOC surpasses that of other baselines, with good diversity. In contrast, the PF obtained by MESMOC and NSGAIII exhibits poor convergence and lacks diversity. NSGAII also yields a dense and diverse PF, albeit with lower convergence compared to VTSMOC. Additionally, its Pareto solutions are mostly dominated by those of VTSMOC.

Five-Objective Scenario: The five-objective formulation has 5 objectives and 4 constraints. The experimental results are summarized in Fig. 12 and Table IV. The reference point is $\boldsymbol{r} = [200(\mu A), 1000(\mu \text{m}^2), 40(\mu A), 10(\mu A), 5(\mu A)]^T$. For this many-objective case, only VTSMOC and NSGAII successfully discover the feasible region. VTSMOC obtains significantly better-Pareto solutions with fewer observations compared to NSGAII, achieving an 8.3× speedup in the number of circuit simulations and a 2.6× speedup in runtime.

2) ACCIA: The second benchmark circuit is an ACCIA [38] designed in the TSMC 0.18-μm process. The ACCIA primarily comprises the current feedback instrumentation amplifier (CFIA), offset reduction loops (ORLs), and choppers. The circuit schematic and design specifications are depicted in Fig. 11(b). A total of 38 free design variables for transistor lengths and widths are considered for sizing, accounting for

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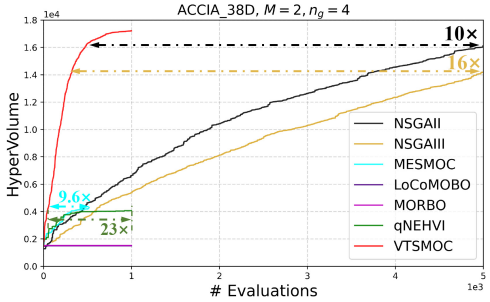


Fig. 14. Optimization results of the ACCIA circuit. The constraints are readily satisfied and the HV curves are plotted.

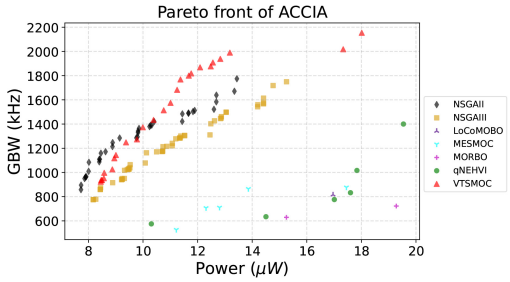


Fig. 15. PF of ACCIA.

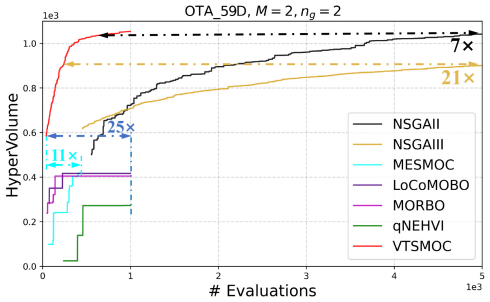


Fig. 16. Optimization results of the OTA circuit. The HV curves are traced from the feasible solutions that satisfy the constraints.

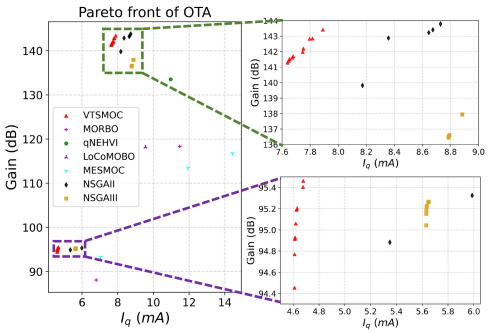


Fig. 17. PF of OTA. OTA's PF exhibits a piecewise continuous structure with two distinct parts. VTSMOC's PF demonstrates superior convergence and diversity compared to other methods across both parts of the PF.

topological symmetry. The optimization metrics consist of 2 objectives and 4 constraints.

The experimental results are presented in Fig. 14 and Table IV. For HV computation, the reference point is set to $\boldsymbol{r} = [20(\mu W), -300(\text{kHz})]$. Due to the ease of satisfying the constraints for ACCIA, we trace the HV curves in Fig. 14 from the solutions that fulfill the constraints. VTSMOC once again outperforms other multiobjective optimization approaches by a substantial margin in terms of both HV improvement and runtime. The trust region-based MOBO methods, LoCoMOBO and MORBO, perform poorly on this benchmark. Compared to the MOEAs NSGAII and NSGAIII, VTSMOC achieves a 10× and 16× speedup, respectively, to reach the same HV with regard to the number of circuit simulations. Additionally, in comparison to the popular qNEHVI method, VTSMOC is 23× faster. The HV curve of MESMOC is incomplete due to its runtime exceeding 50 h. VTSMOC is also 9.6× faster than MESMOC in terms of circuit simulations. When considering computational resource consumption, VTSMOC realizes a 20× runtime speedup over the MOEAs NSGAII and NSGAIII, while also delivering superior results.

Fig. 15 compares the PFs of ACCIA obtained by different approaches. The ACCIA exhibits a linear PF. VTSMOC, NSGAII, and NSGAIII all achieve high-quality PFs with good convergence and diversity. NSGAII's PF has a slight advantage in the low-power area with Power $\leq 8\mu W$, while VTSMOC's PF dominates the high-gain-bandwidth product (GBW) area with GBW $\geq$ 1800 kHz. The PFs obtained by MORBO and LoCoMOBO exhibit poor convergence and contain few observations.

3) OTA: Another circuit is a low-power, high-efficiency, high-swing OTA, employing a two-stage structure with a hybrid cascode compensation scheme [39], [40]. The circuit schematic and optimization metrics are illustrated in Fig. 11(c). The OTA incorporates components, such as the two-stage amplifier, negative-side cascode, positive-side cascode, and the bias circuit, totaling 71 transistors, four capacitors, and two resistors. The circuit is designed in TSMC 0.18-μm process. The formulated optimization task comprises $D = 59$ free design variables, $M = 2$ objectives, and $n_g = 2$ constraints.

The experimental results are presented in Table IV and Fig. 16. The reference point is chosen as $\boldsymbol{r} = [15(mA), -20(\text{dB})]$ for HV computation. Since the constraints for the OTA circuit are easy to satisfy, we only track the HV curves from solutions that meet the constraints in Fig. 16. Consistently, VTSMOC exhibits superior performance, characterized by markedly higher-sample efficiency and lower-computational cost. Compared to MOEAs, VTSMOC, respectively, achieves a 7× and 21× speedup over NSGAII and NSGAIII, to obtain the same HV results when considering the number of circuit simulations. The performance of MOBO baselines is poor on this high-dimensional CMOO benchmark. The curve of MESMOC is incomplete as its runtime exceeds 50 h. Although MESMOC is prohibitively slow, it is more successful in finding the feasible region compared to other HVI-based MOBO baselines. VTSMOC is 25× faster than HVI-based approaches, such as MORBO, qNEHVI, and LoCoMOBO, and 11× faster than MESMOC. When it comes to runtime, VTSMOC also showcases a prominent advantage. VTSMOC achieves a 9.4× runtime speed up than NSGAII with comparable results.

Fig. 17 displays a comparison of the obtained PFs using different approaches. The PF of OTA has a piecewise continuous structure with two distinct parts. Both the proposed VTSMOC and the MOEA methods, NSGAII and NSGAIII, demonstrate

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TABLE V
COMPARISON OF EXPERIMENTAL RESULTS WITH VTS [21]

Algorithms		ChargePump_M2	ChargePump_M5	ACCIA	OTA
VTSMOC	HV	**1.81±0.06e5**	**4.90±0.63e7**	**1.72±0.08e4**	**1.05±0.04e3**
	FoM	**1.24±0.04e2**	6.84±0.60e2	**3.43±0.10e2**	39.6±5.9
	Runtime	**0.46h**	**0.97h**	**1.2h**	**0.70h**
VTS [21]	HV	1.78±0.08e5	3.61±0.64e7	8.53±2.72e3	8.48±0.96e2
	FoM	1.28±0.05e2	**6.07±0.52e2**	4.31±0.34e2	**39.4±2.6**
	Runtime	1.2h	1.1h	5.3h	1.9h

considerable convergence across both parts of the PF. Among these methods, VTSMOC exhibits the best convergence and diversity, containing the highest number of Pareto solutions within the PF. However, the PF obtained by other MOBO approaches is very sparse, containing only a few solutions.

D. Improvements to VTS

VTSMOC improves VTS [21] primarily from two aspects: 1) introducing new features to tackle MOO problems and 2) incorporating parallel sampling strategies. By separately modeling each objective and constraint, VTSMOC can capture the circuit behaviors more effectively and achieves superior Pareto solutions. The performance comparison of VTSMOC and VTS [21] with a simulation budget of 1000 is presented in Table V. For the SOO method VTS, we optimize the Figure of Merit (FoM) defined as follows:

$$\text{FoM}(\boldsymbol{x}|\boldsymbol{\lambda}) = \boldsymbol{\lambda}^T \boldsymbol{f}(\boldsymbol{x}) + \text{CV}(\boldsymbol{x}) \tag{27}$$

where $\boldsymbol{\lambda} = [\lambda_1, \ldots, \lambda_M]^T$ is an empirical scalarization factor with $\lambda_i > 0$ for all $i \in \{1, \ldots, M\}$. Hence, the optimal FoM for the SOO corresponds to the minimal projection value of the PF for the MOO problem under the given scalarization vector $\boldsymbol{\lambda}$. For all benchmark circuits, VTSMOC persists to exhibit a significant advantage in both HV and runtime, demonstrating its superiority in discovering a diverse PF. Benefit from the parallel sampling strategy, VTSMOC still achieves shorter runtime despite the additional computational cost associated with building multiple GP models. Additionally, VTSMOC obtains comparable FoM values with VTS under the same simulation budget. In contrast, the SOO method VTS only attains satisfactory FoM values in the given direction $\boldsymbol{\lambda}$ and performs poorly in other directions, resulting in a lower HV.

VI. CONCLUSION

We introduce VTSMOC, a sample-efficient and computationally lightweight BO framework tailored for high-dimensional CMOO problems. By employing dynamic Voronoi tree construction, we limit the maximum observations in each leaf node, effectively reducing computation costs. The dispersed batch selection strategy in the objective space of promising Voronoi cells ensures both diversity and quality of the PF. To prevent blind exploration, local MOBO is conducted within the selected promising cells. Additionally, we propose the EPFI and PPFI multiobjective sampling criteria to efficiently facilitate the PF surface along the radial direction. Due to their explicit formulations, EPFI and PPFI are computationally inexpensive and readily extendable to many-objective and constrained scenarios. Experimental results demonstrate VTSMOC's significant improvements in sample and computational efficiency compared to state-of-the-art MOO approaches.

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